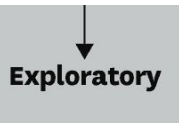


introduction to

DATA DATA DATA
VISUALISATION

#1



basics

Whats Is dataviz ?
Why do we visualize ?
Visual Functions

dataviz for data exploration

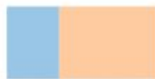
Search for
meaning using
visualisation



dataviz for communication

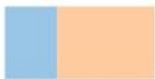
Show findings
using visualisation

Class & TP #1



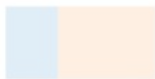
28 & 29 march

Class & TP #2



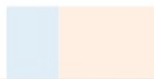
14 & 19 / 20 april

Class & TP #3



5 & 9 may

Class & TP #4



19 & 23 may



LAST DAYS

basics

dataviz for data exploration

- I. Whats is it ?
- II. Why do we visualize ?
- III. Visualisation Functions
- IV. Starting to explore with datavis

- I. Quizz
- II. Chart of the day
- III. Try dashboards
- IV. About dashboards
- V. Let's explore again (plotly)
- VI. Exploration for others :
dashboard

↓
Exploratory

Declarative
↑

basics

Whats Is dataviz ?
Why do we visualize ?
Visual Functions

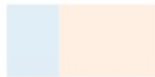
**dataviz for data
exploration**

Search for
meaning using
visualisation

**dataviz for
communication**

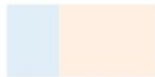
Show findings
using visualisation

Class & TP #1



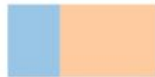
28 & 29 march

Class & TP #2



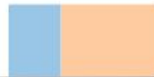
14 & 19 / 20 april

Class & TP #3



5 & 9 may

Class & TP #4



19 & 23 may



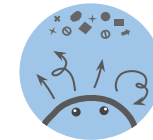
TODAY

Show findings using visualisation (½)

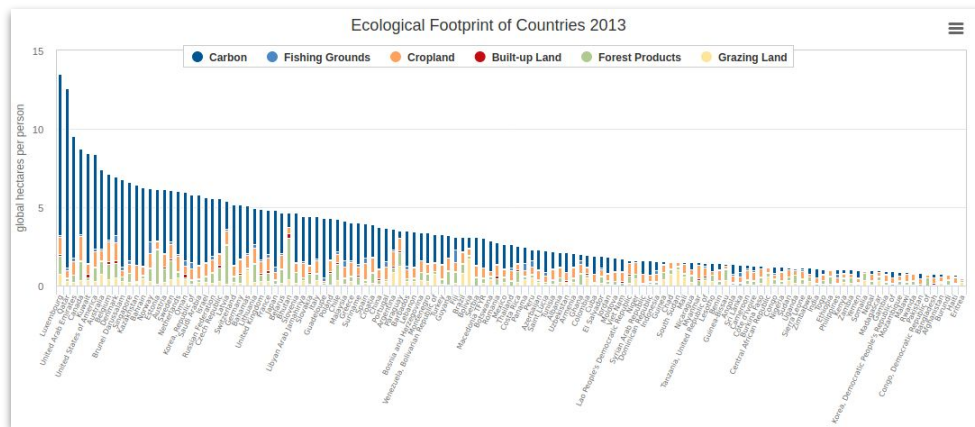
- I. Chart of the day
- II. Datavis for communication basics

- I. Chart of the day
- II. Datavis for communication basics

Objectives



- What works and what doesn't work with this chart?
- How can you make it better?



- **Original goal** : compare countries footprint per capita with footprint details
- **But more data available** : biocapacity per capita and the total footprint of a country
- **Let's reduce the information and present it in a novel way** : are countries debtors or creditors : footprint VS biocapacity

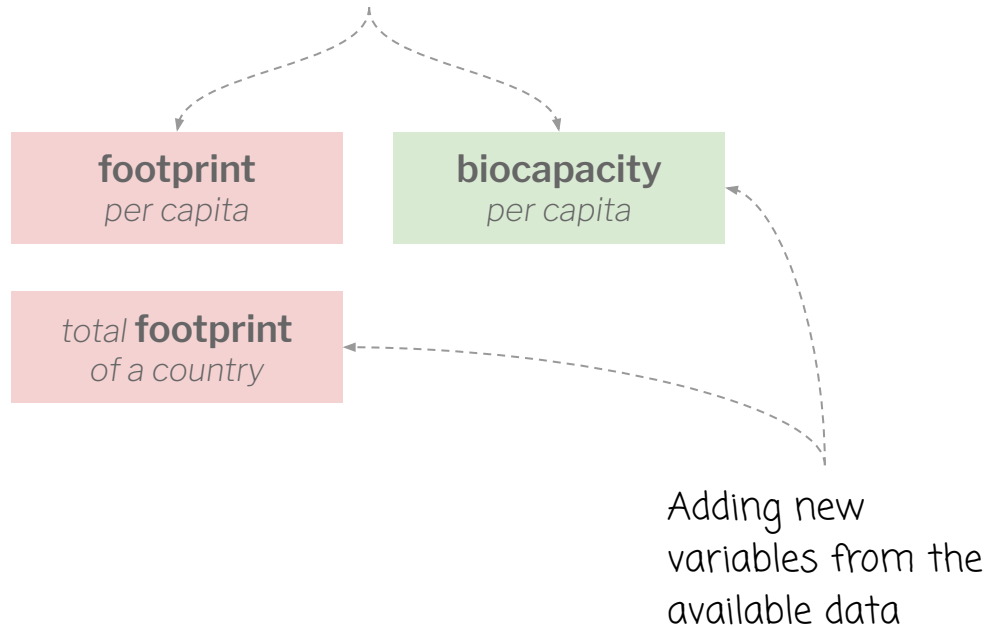
Ecological Footprint is a measure of how much area of biologically productive land and water an individual, population, or activity requires to produce all the resources it consumes and to absorb the waste it generates, using prevailing technology and resource management practices.



#3

→ Let's reduce the information and present it in a novel way : are countries debtors or creditors

Comparing those 2 as the new main message of the chart

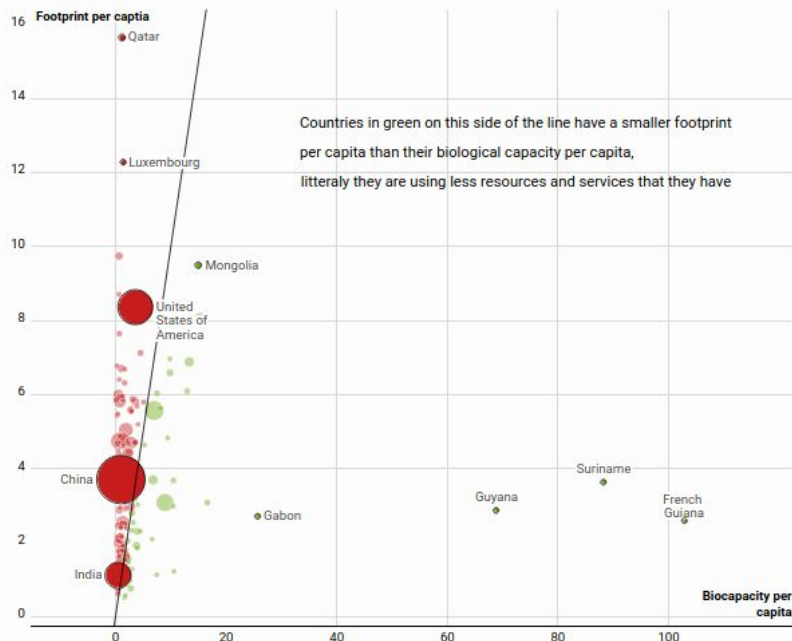




#3

Debitors or creditors of the world ? Looking at countries ecological footprint versus biocapacity

Size of the circle represent the total Ecological Footprint of the country. Beware that the countries with the biggest footprint (China, USA, India ...) might be comparatively small looking at the Footprint per capita. Qatar and Luxembourg are way ahead while using more resources than they have !



Based on 2014 data. Gha or Global hectares are the accounting unit for the Ecological Footprint and biocapacity accounts. Know more in 3 minutes : <https://www.youtube.com/watch?v=TSM3MIPW4&feature=youtu.be>

Chart: Made by Edith Visualize • Source: [Global Footprint Network](#) • [Get the data](#) • Created with [Datawrapper](#)

→ The two main variables - footprint and biocapacity - are in fact one categorical one :

- ~ Debtors (footprint > biocapacity)
- ~ Creditors (biocapacity > footprint)

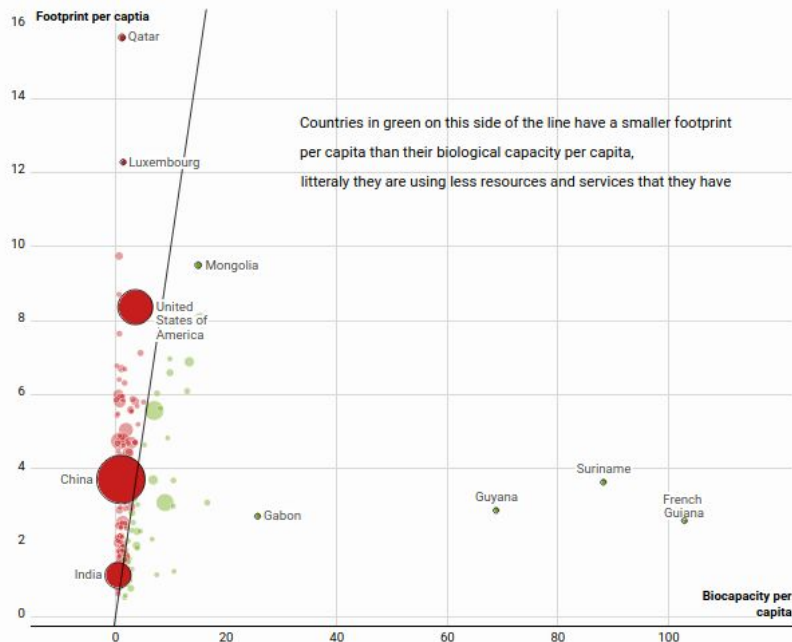
Of course this is an approximation because we are talking about "per capita" values



#3

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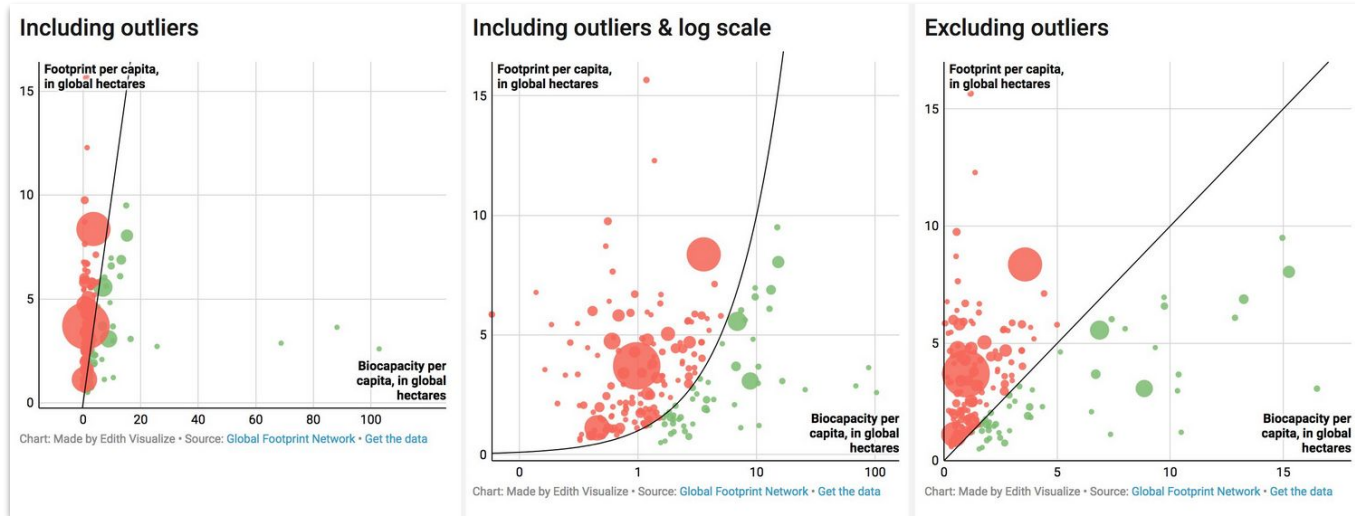
Of course this is an approximation because we are talking about "per capita" values

→ It's the most important, we want to make it obvious : we can double encode it

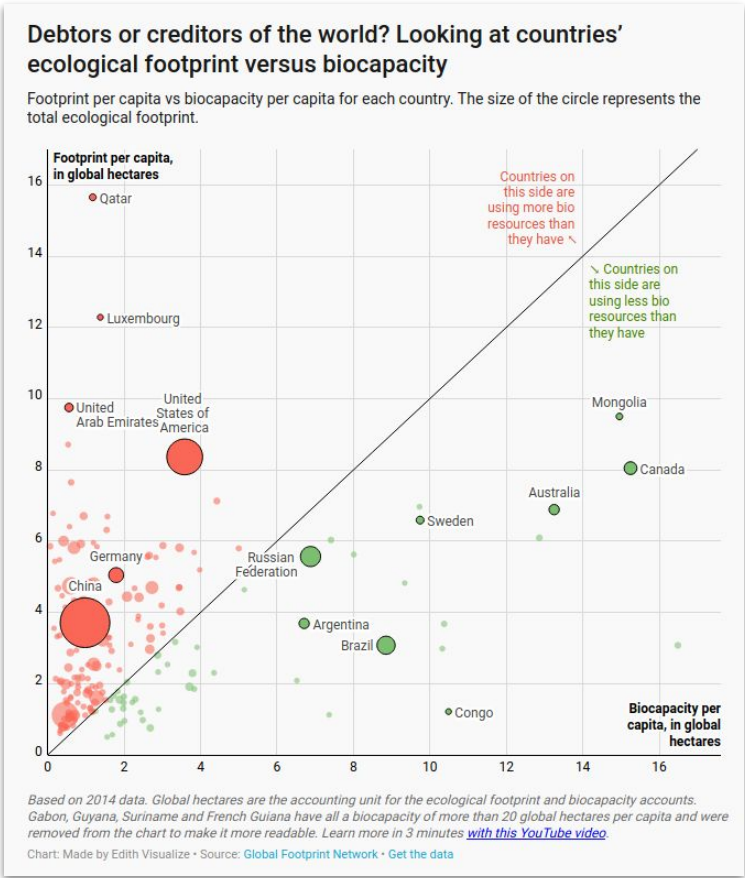
- ~ Position (x,y) 2 triangles
- ~ Color



Making the graph more lisible
Changing axis type, excluding outliers



by [Lisa Rost](#)



More lisible annotations

More detailed axis legend

by [Lisa Rost](#)

- I. Chart of the day
- II. Datavis for communication basics



How to make a data visualisation for others ? #present #communication



encoding

goals

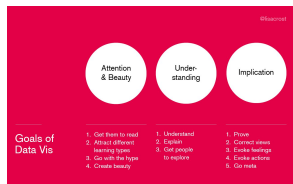
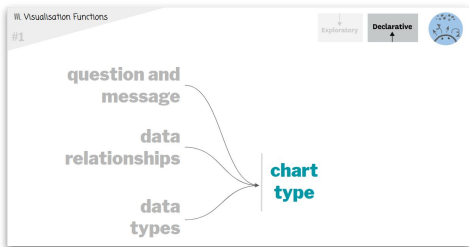
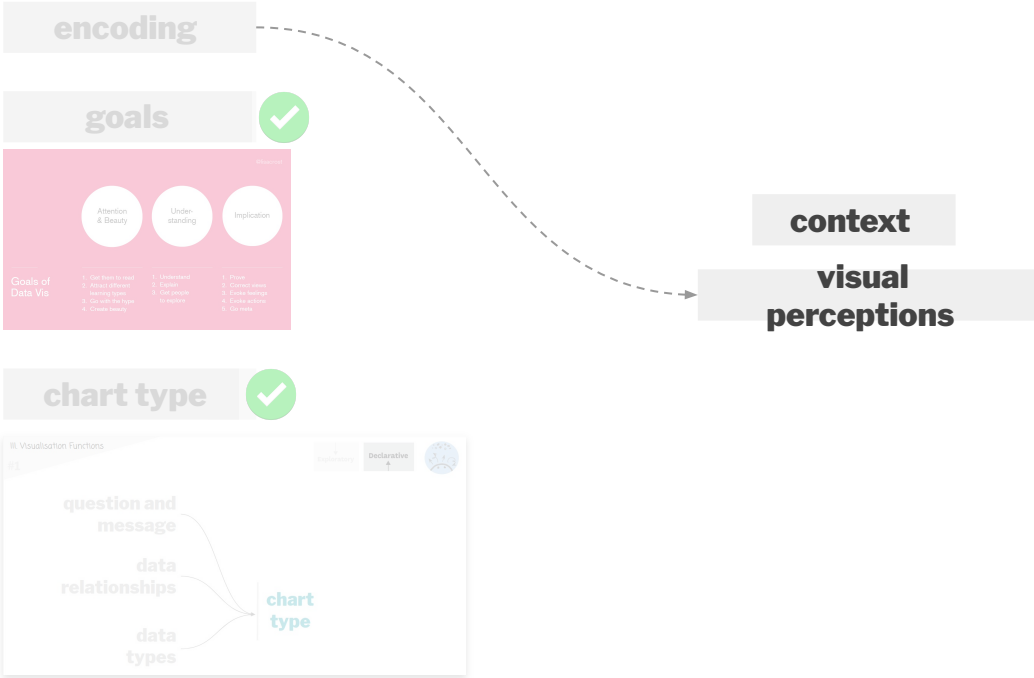


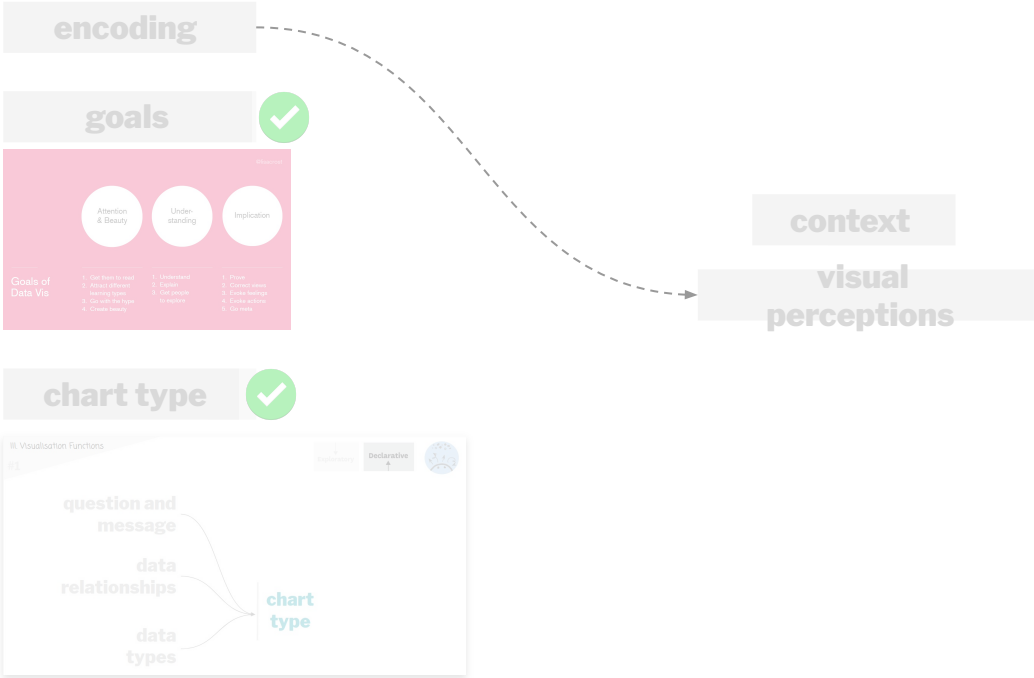
chart type



How to make a
data
visualisation for
others ?
#present
#communication



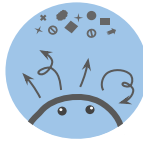
How to make a
data
visualisation for
others ?
#present
#communication



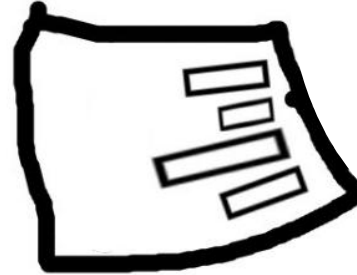
How to make a data visualisation for others ?

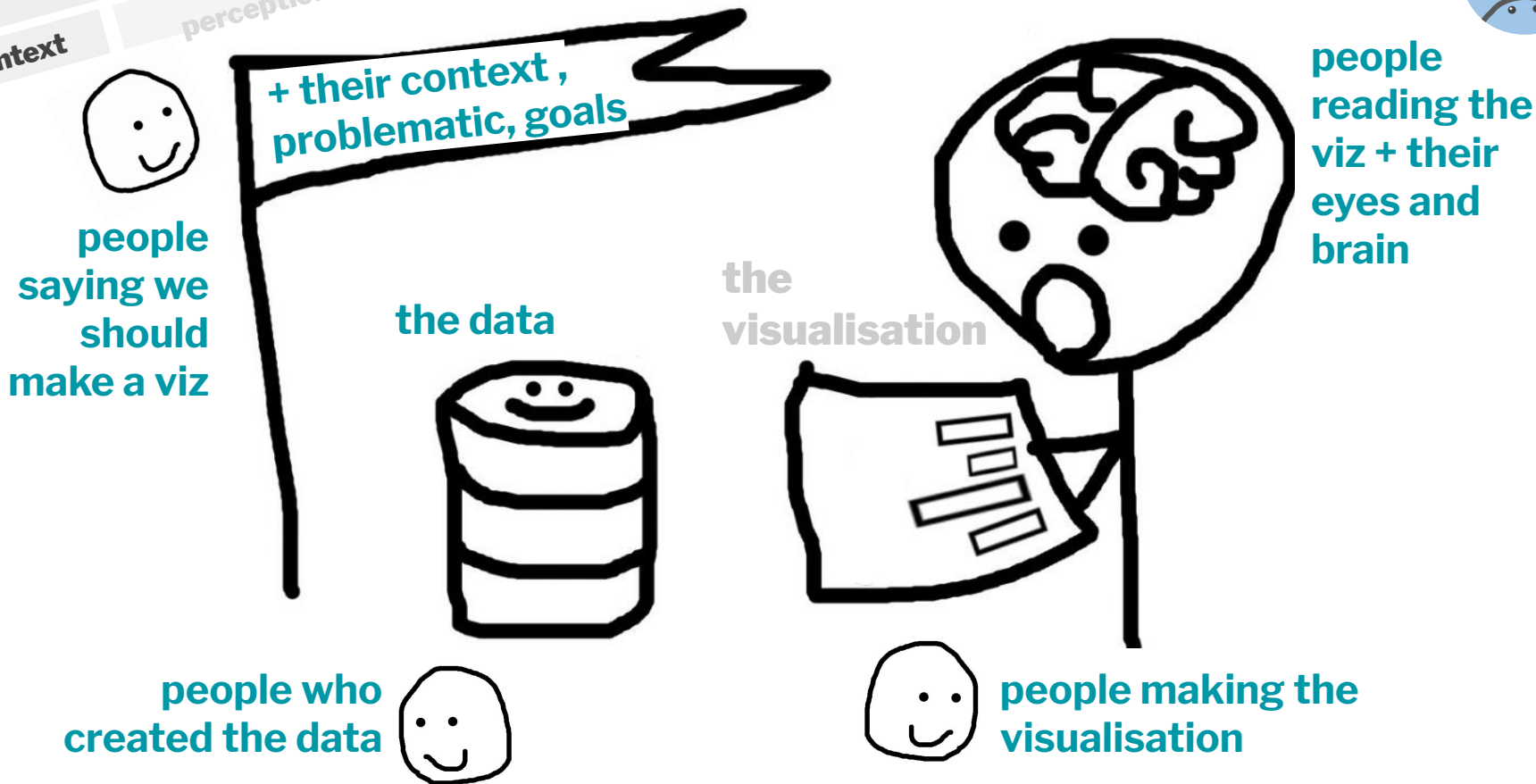
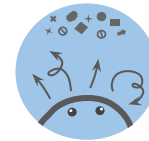
**datavis
framework**

**best
practices**



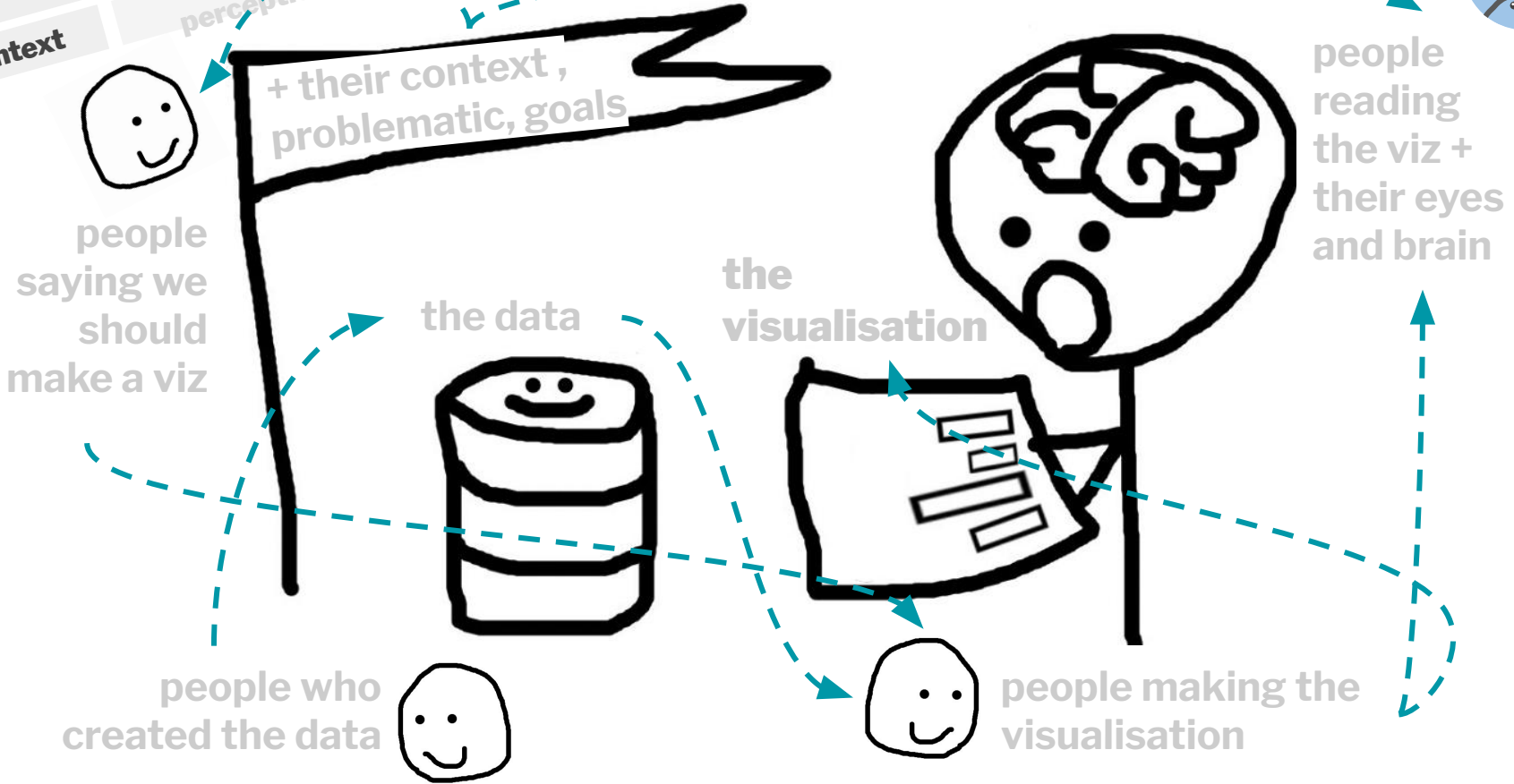
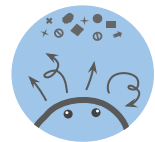
the visualisation





context

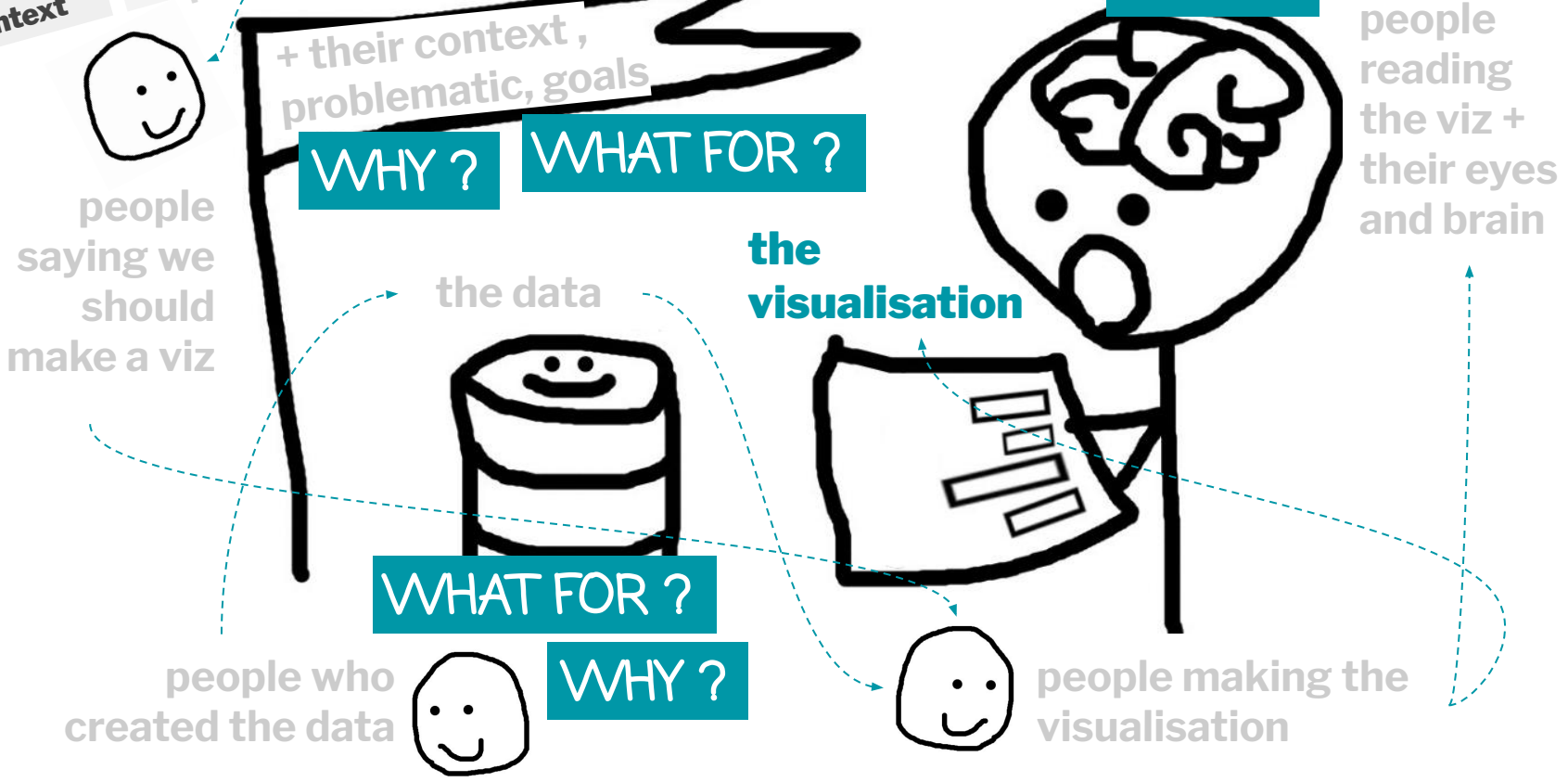
visual perceptions



#3

context

visual perceptions





the data



WHAT FOR ?

people who
created the data



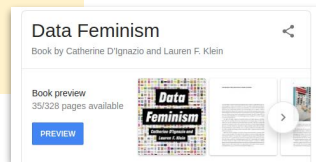
WHY ?

Out of our scope

But always be reminded that data is a proxy for reality, we should question it.

Who collected it ? How ? In what purpose ? Is there interests ? Is the collection method consistent ? How can you check if it is close to reality, how does it really look like on the ground ?

Are there other data sources complementary, or more aligned with our goals ? Are there things that are not in the data ?





#3

context

visual
perceptions



people
saying we
should
make a viz

+ their context,
problematic, goals

WHY ?

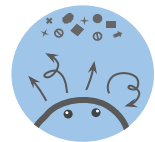
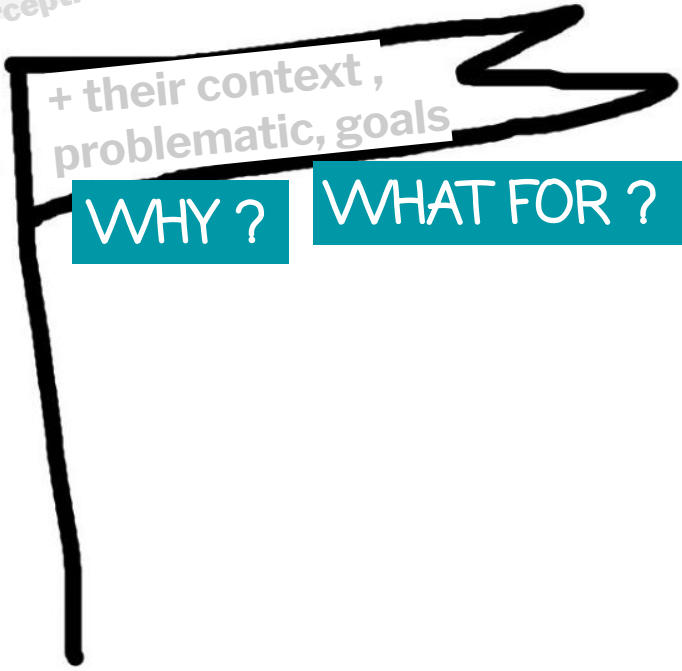
WHAT FOR ?

context

visual perceptions



people
saying we
should
make a viz



@lisacrost

	Attention & Beauty	Under- standing	Implication
Goals of Data Vis	1. Get them to read 2. Attract different learning types 3. Go with the hype 4. Create beauty	1. Understand 2. Explain 3. Get people to explore	1. Prove 2. Correct views 3. Evoke feelings 4. Evoke actions 5. Go meta



people
reading
the viz +
their eyes
and brain



Is there only one type of audience ?

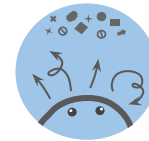
How often do they see / use data
visualisation ? What would be the chart
types they know ?

What do they need visualisation for ?

**KNOW YOUR
AUDIENCE**

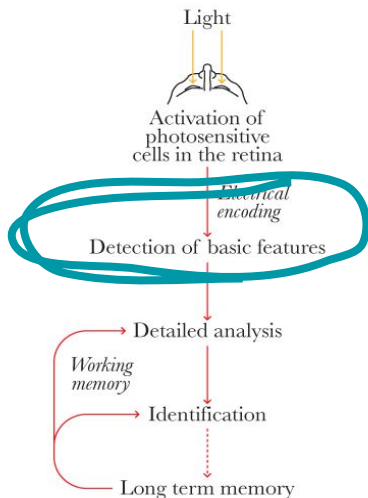


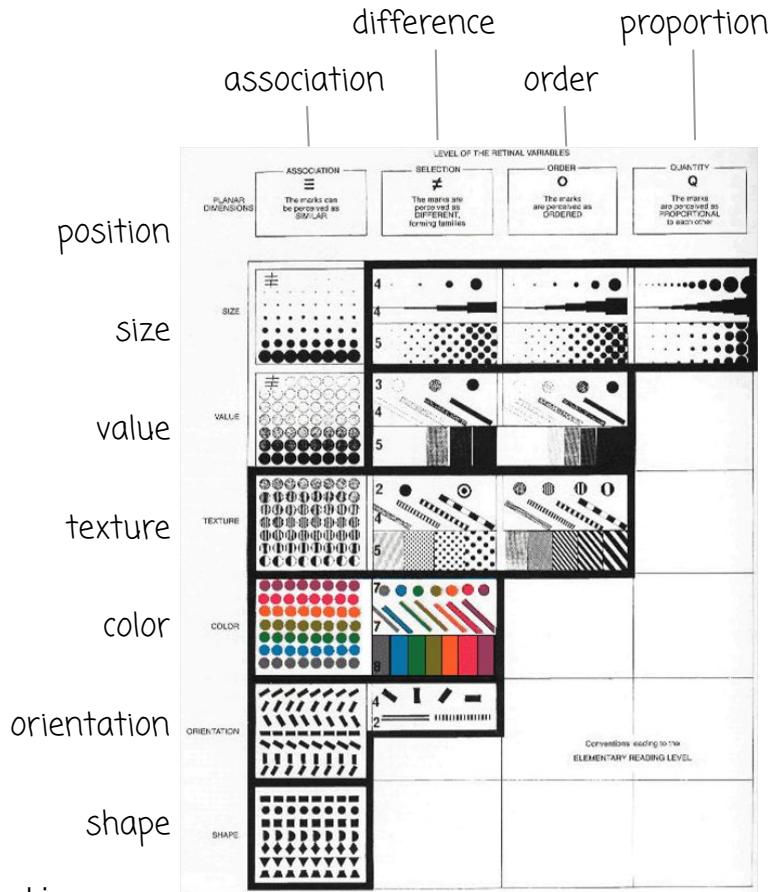
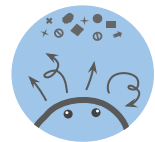
people
reading
the viz +
**their eyes
and brain**



people
reading
the viz +
their eyes
and brain

preattentive features of our brain

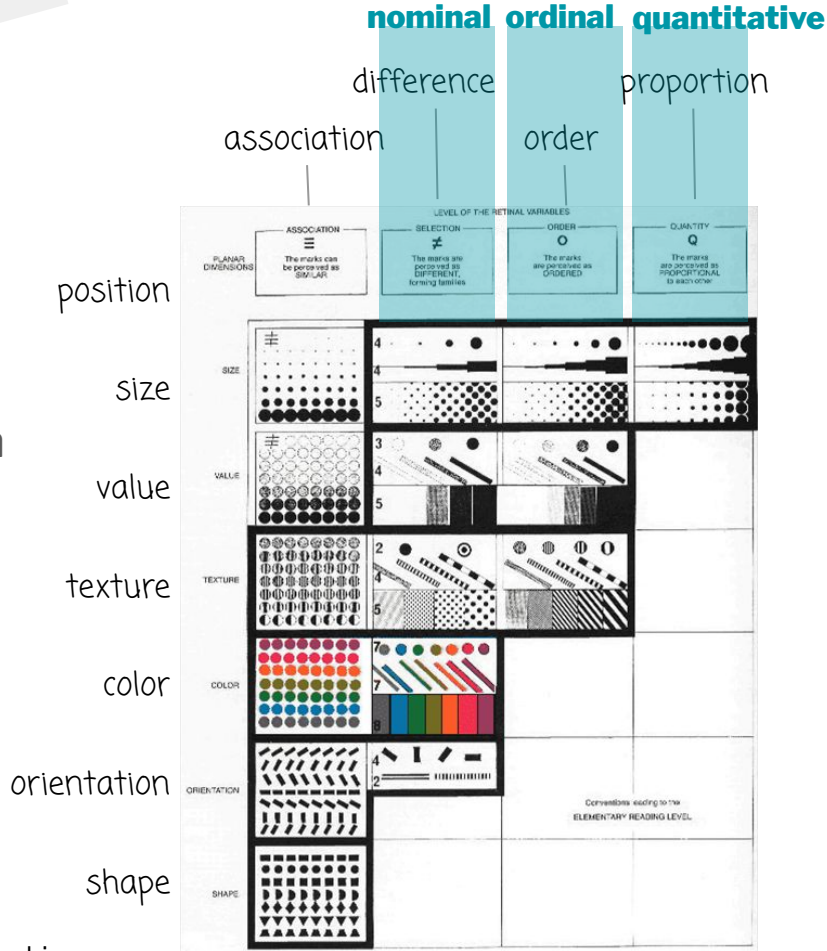






CHOOSING ENCODINGS

Consistency between variable type and data type





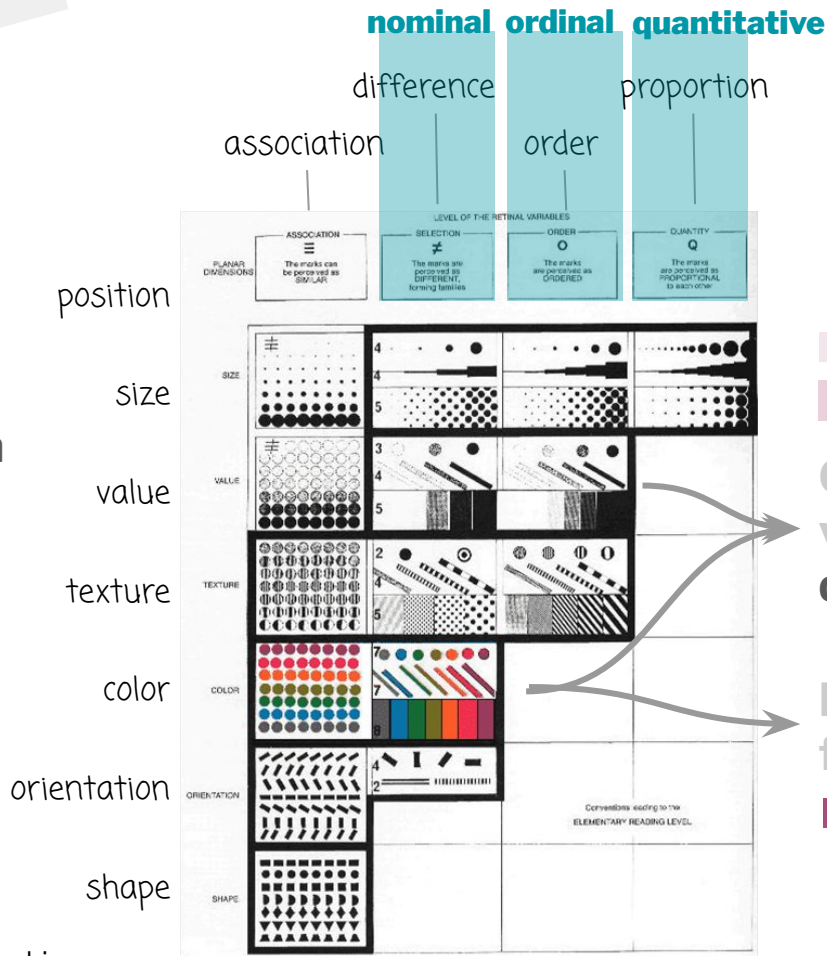
#3

visual
perceptions

context

**CHOOSING
ENCODINGS**

**Consistency between
variable type and
data type**



One color and different
values for ordinal &
quantitative variables

Different colors (hue)
for nominal variables

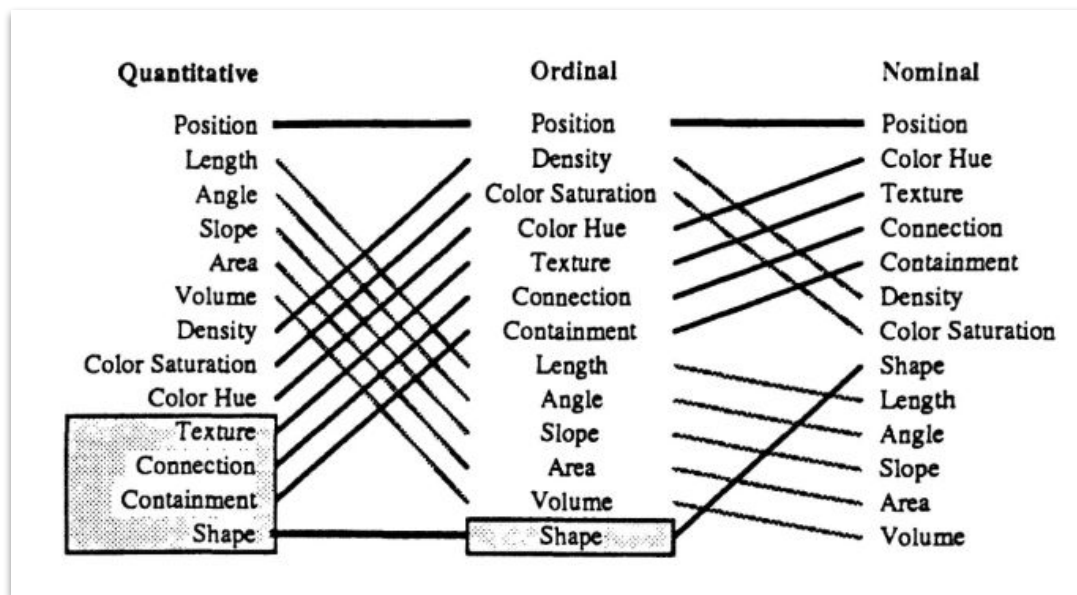




CHOOSING ENCODINGS

Consistency between
variable type and
data type

The most important
variable, the most
effective encoding

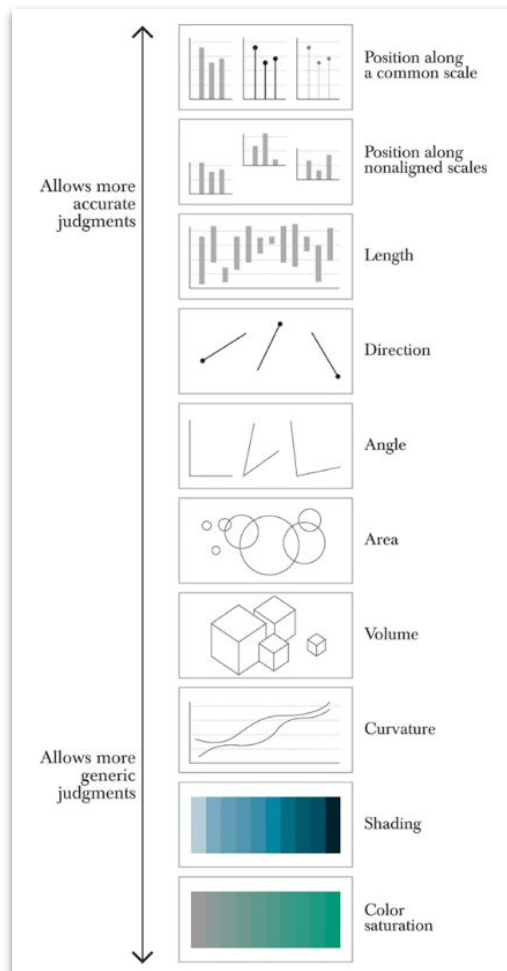


CHOOSING ENCODINGS

Consistency between
variable type and
data type

The most important
variable, the most
effective encoding

**Adapt to accurate or
generic judgment**

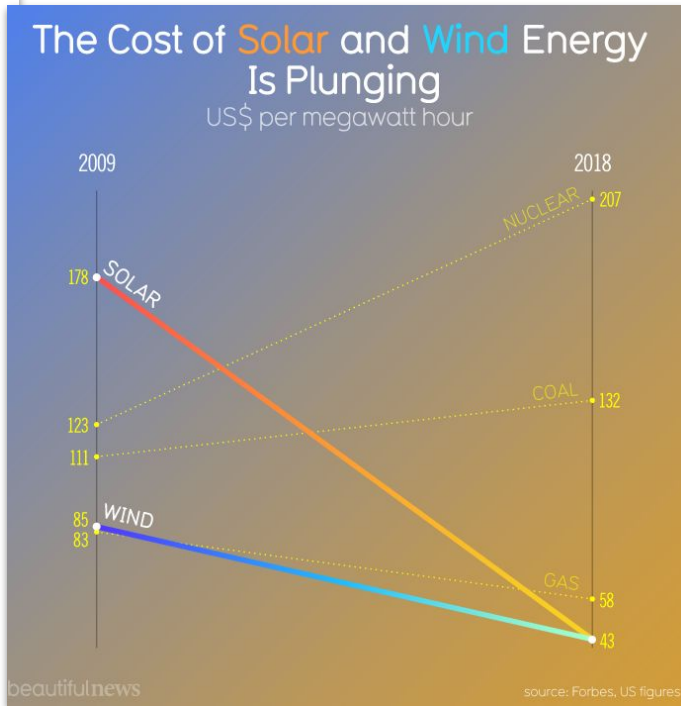
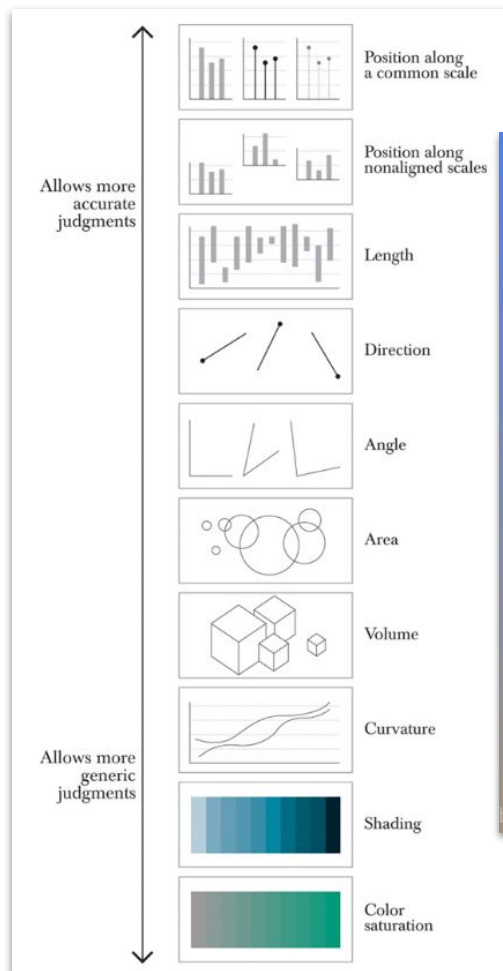


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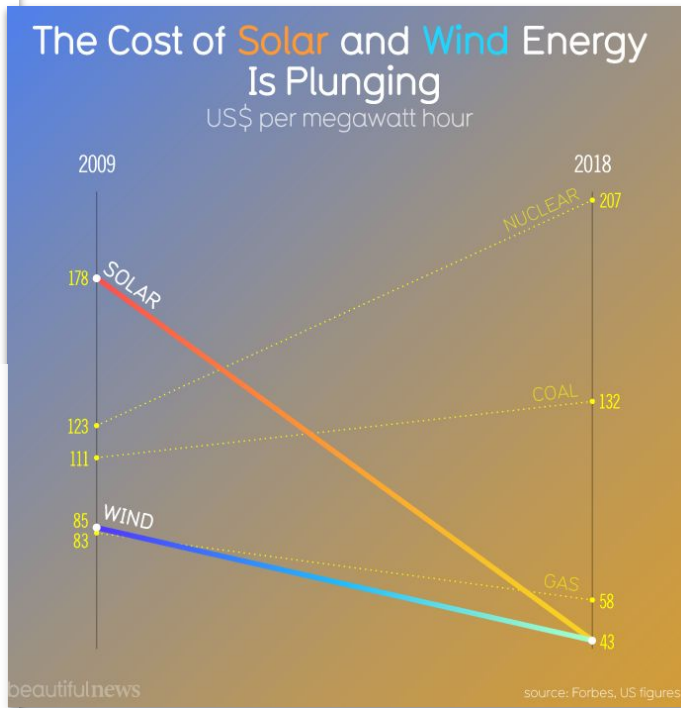
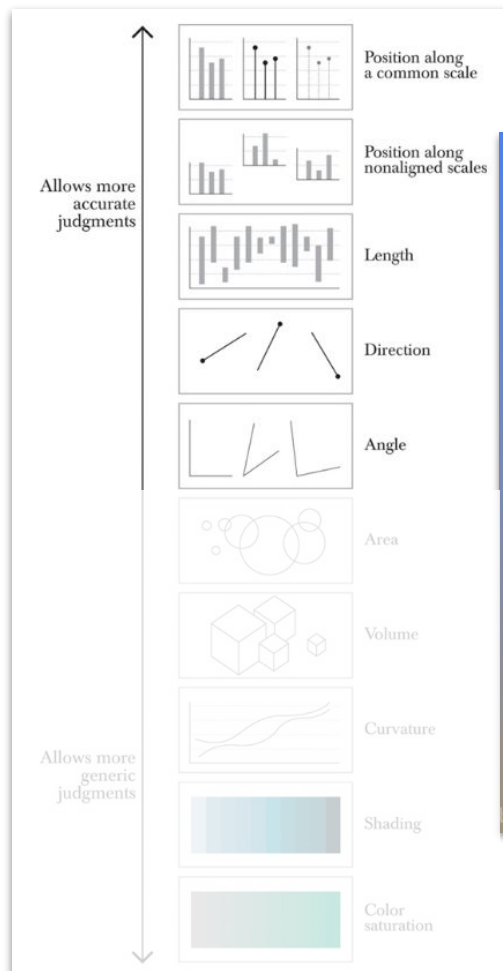


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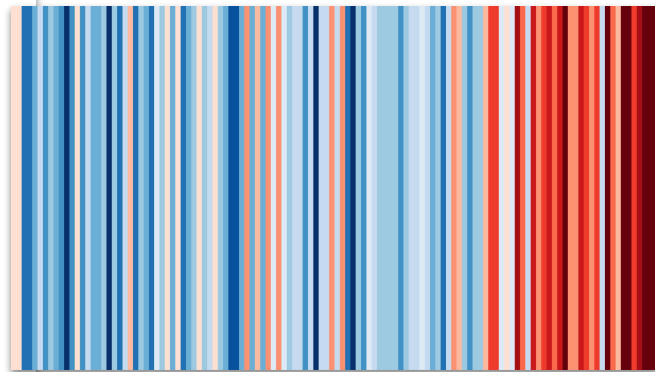
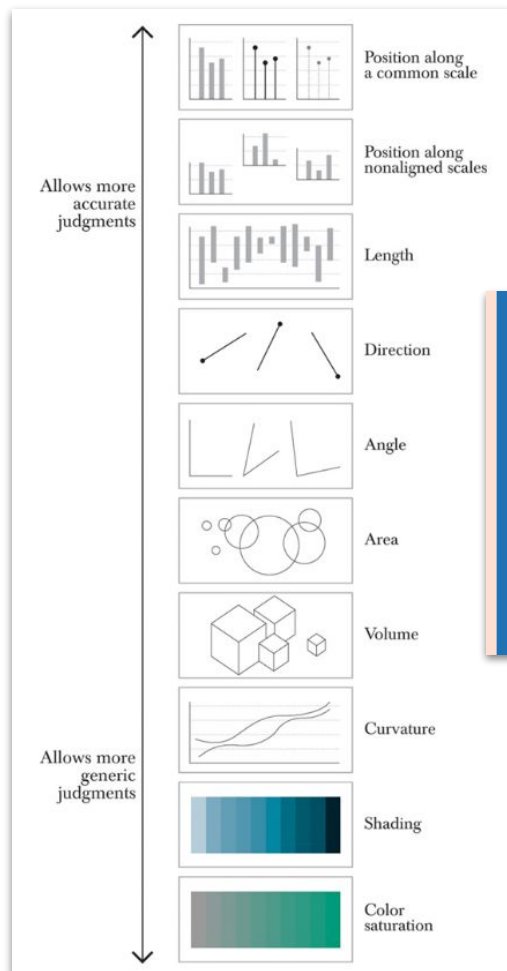


CHOOSING ENCODINGS

Consistency between
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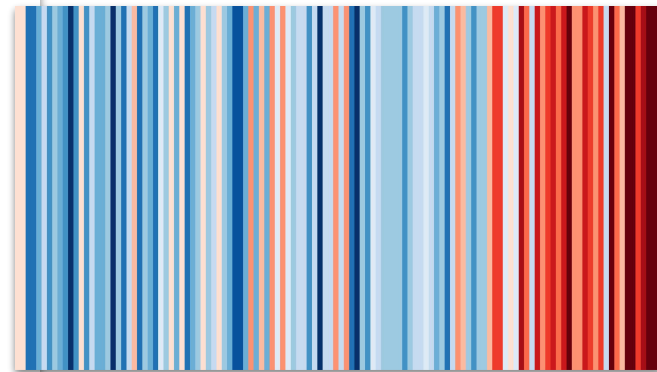
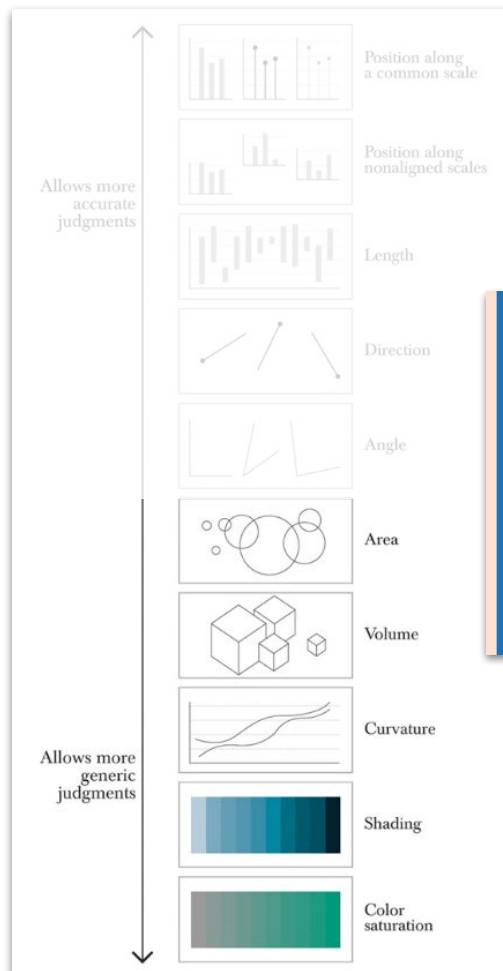


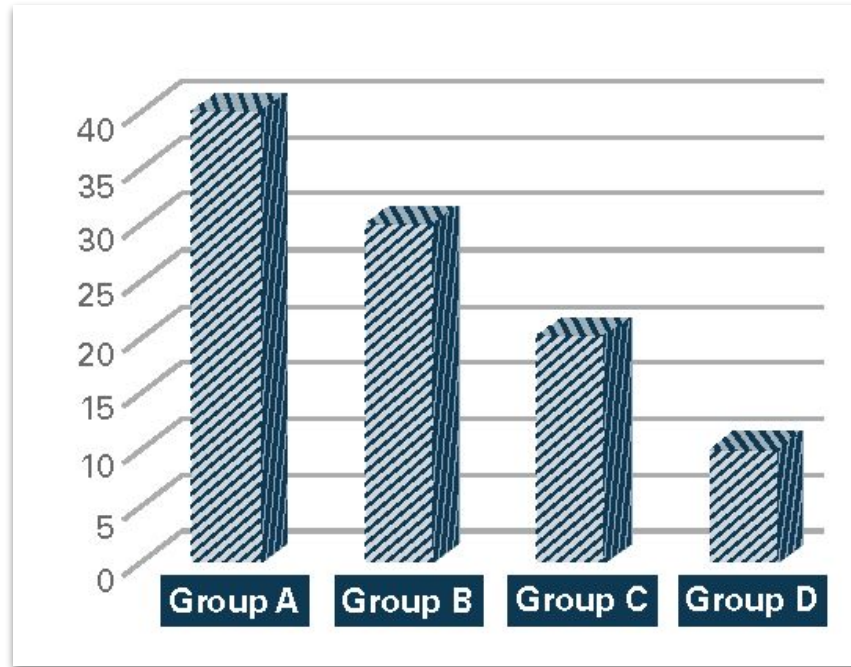
CHOOSING ENCODINGS

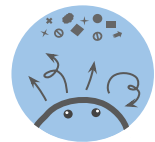
Consistency between
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The most important
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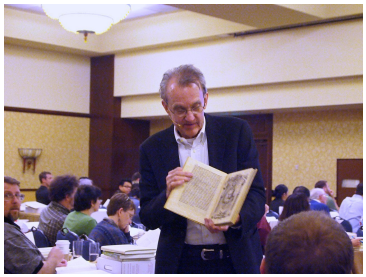
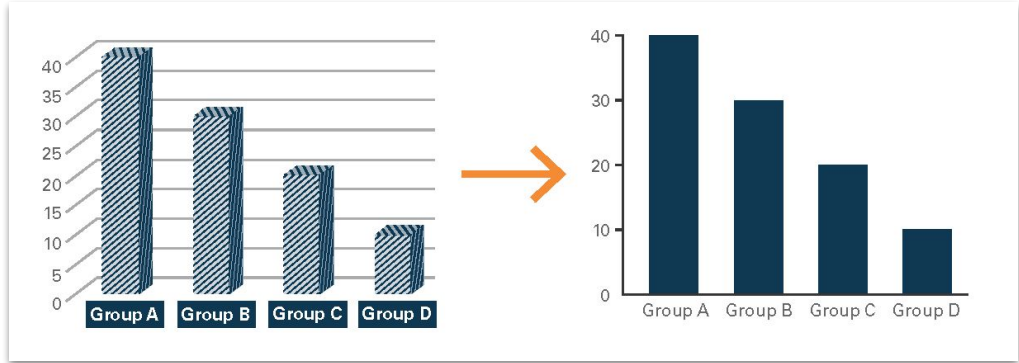
**Adapt to accurate or
generic judgment**







data ink ratio



Data-ink ratio =
$$\frac{\text{Data-ink}}{\text{Total ink used to print the graphic}}$$

= proportion of a graphic's ink devoted to the non-redundant display of data-information

= 1.0 - proportion of a graphic that can be erased

**LESS
IS MORE**

gestalt principles



Law of proximity



0:01 | 55 k vues

4 cards

Common Region



0:01 | 8,6 k vues

Grouped elements

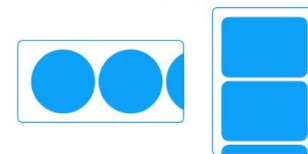
Similarity Principle



0:01 | 9,9 k vues

3 groups

Closure Principle



0:01 | 8,8 k vues

Scrollable

Focal Point



0:01 | 9,2 k vues

Focus 🙌

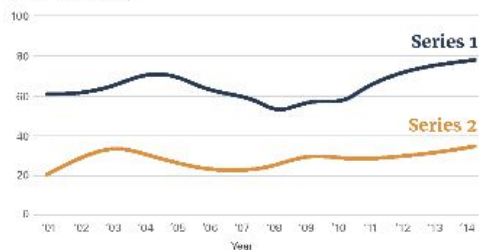
Figure-Ground Articulation



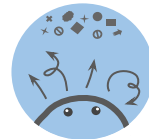
Foreground and background

Chart Title Here

(Y axis label here)

LEGEND
CLOSE TO DATASTRUCTURE
INFORMATIONUSE
PADDING

gestalt principles



Law of proximity



0:01 | 55 k vues

4 cards

Common Region



0:01 | 8.6 k vues

Grouped elements

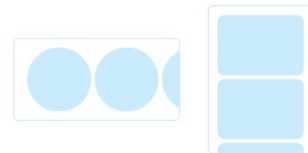
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0:01 | 9.9 k vues

3 groups

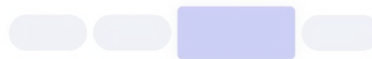
Closure Principle



0:01 | 8.8 k vues

Scrollable

Focal Point



0:01 | 9.2 k vues

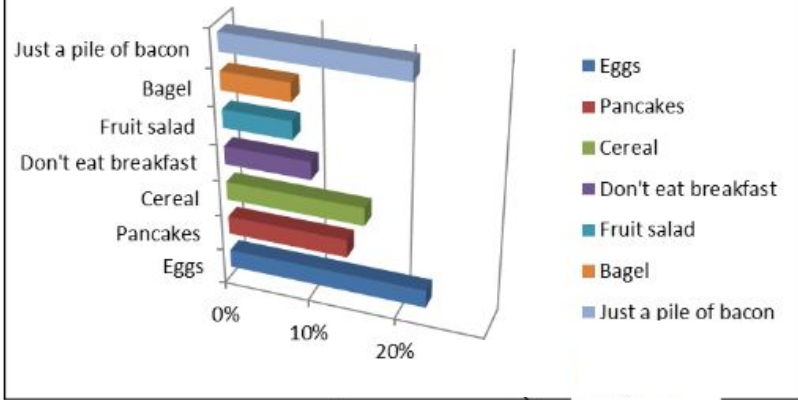
Focus 📌

Figure-Ground Articulation

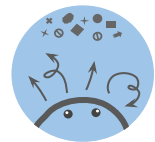


Foreground and background

Attendee Breakfast Preferences



gestalt principles



Law of proximity

0:01 55 k vues

Common Region

0:01 8.6 k vues

Grouped elements

Similarity Principle

0:01 9.9 k vues

3 groups

Closure Principle

0:01 8.8 k vues

Scrollable

Focal Point

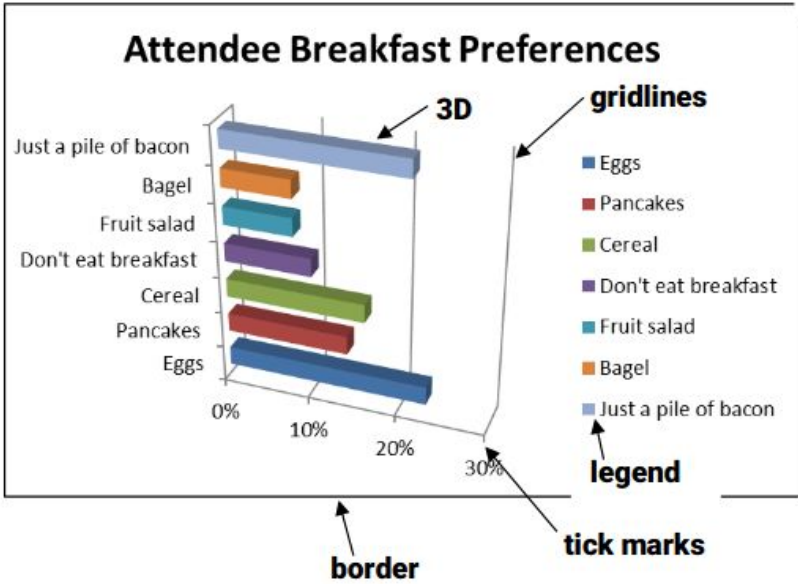
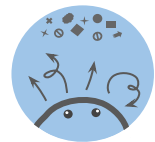
0:01 9.2 k vues

Focus

Figure-Ground Articulation

Foreground and background

gestalt principles



Law of proximity

0:01 55 k vues

Common Region

0:01 8.6 k vues

Similarity Principle

0:01 9.9 k vues

Closure Principle

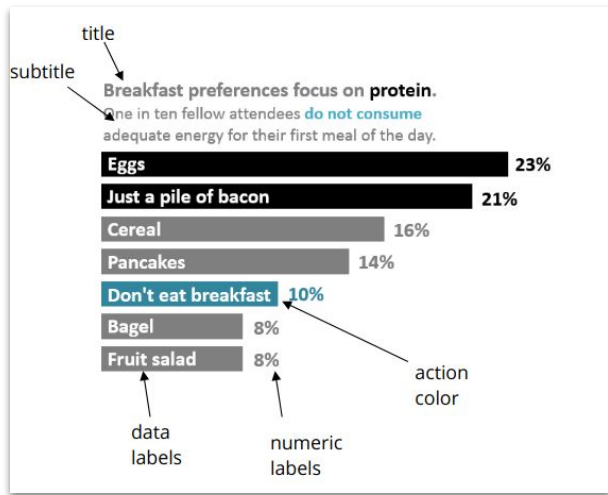
0:01 8.8 k vues

Focal Point

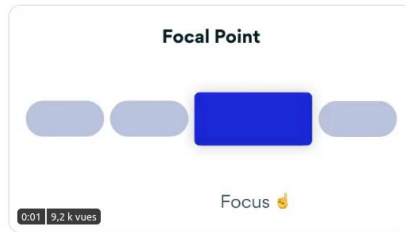
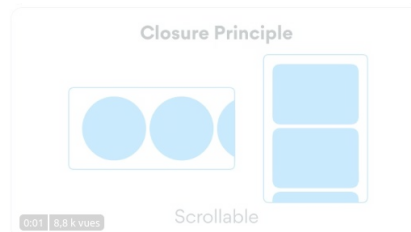
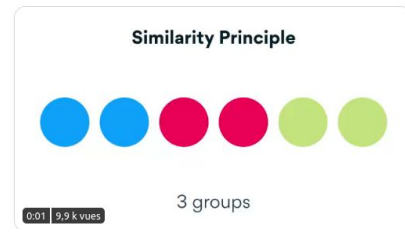
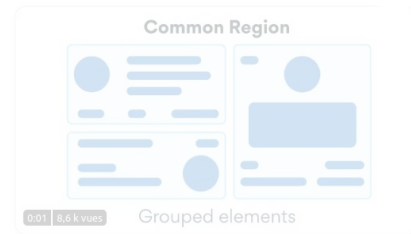
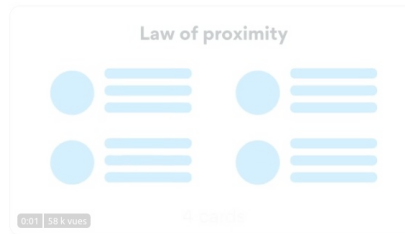
0:01 9.2 k vues

Figure-Ground Articulation

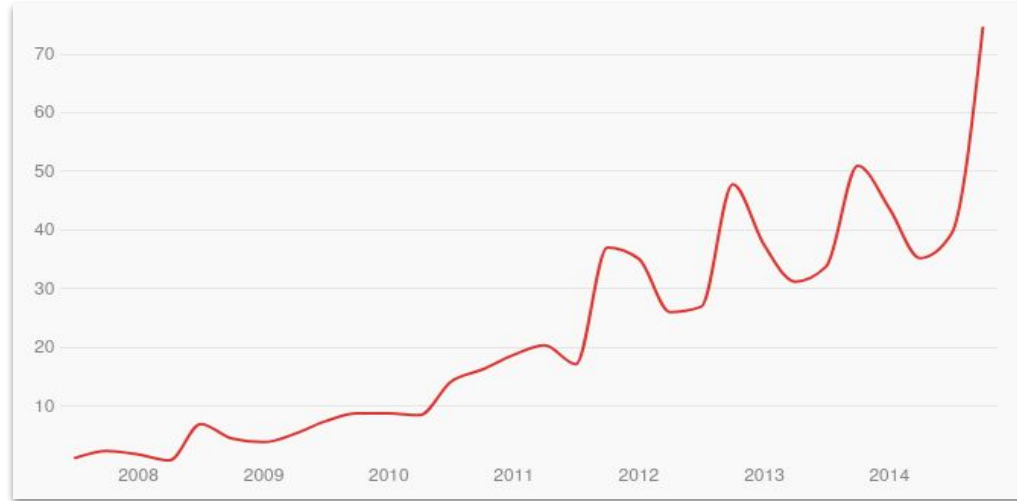
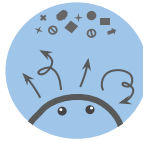
0:01 9.2 k vues

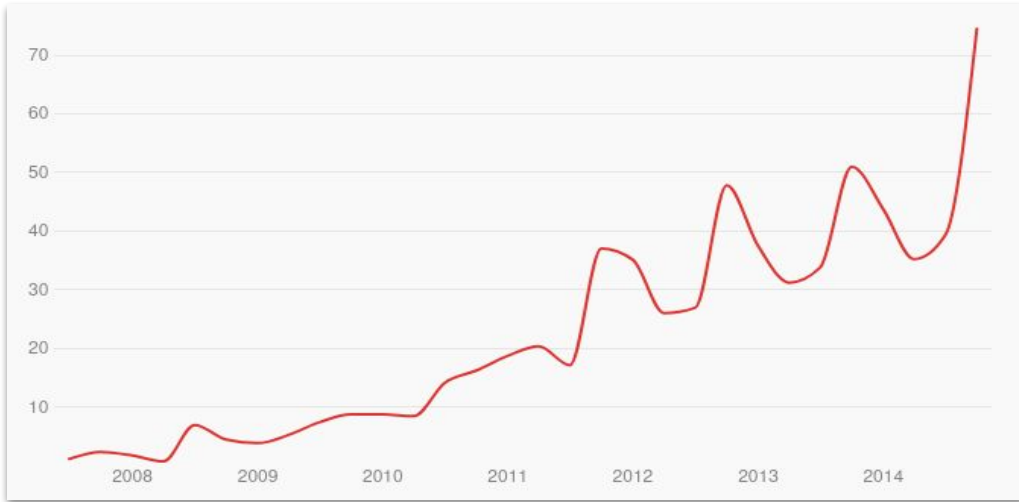
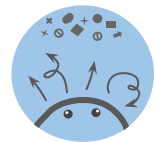


gestalt principles

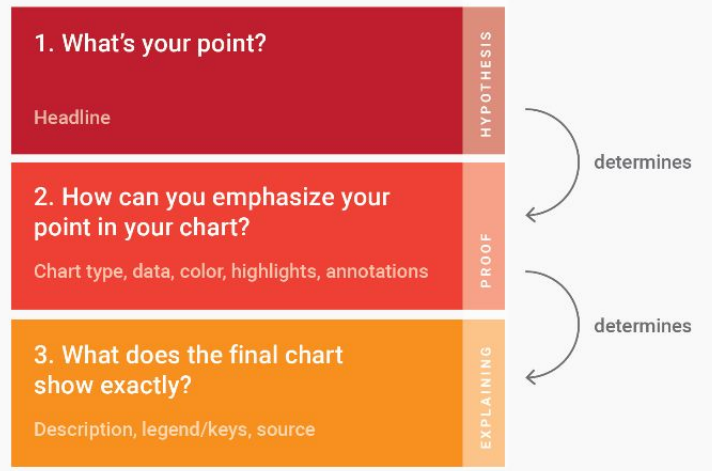
CONSISTENT
COLOR CODING

HIGHLIGHT





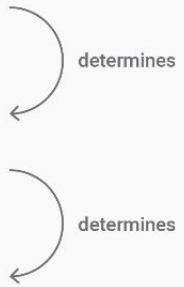
Three questions for creating a chart



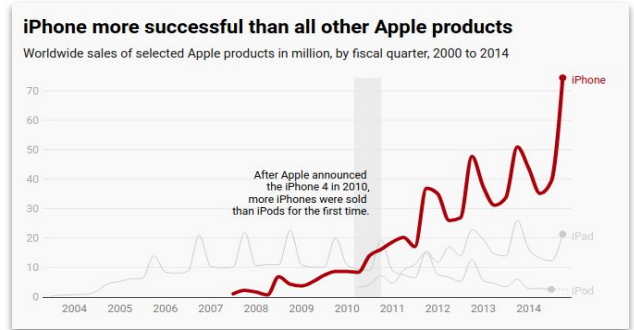
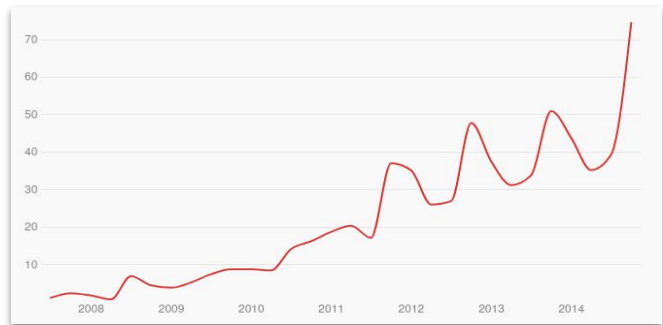


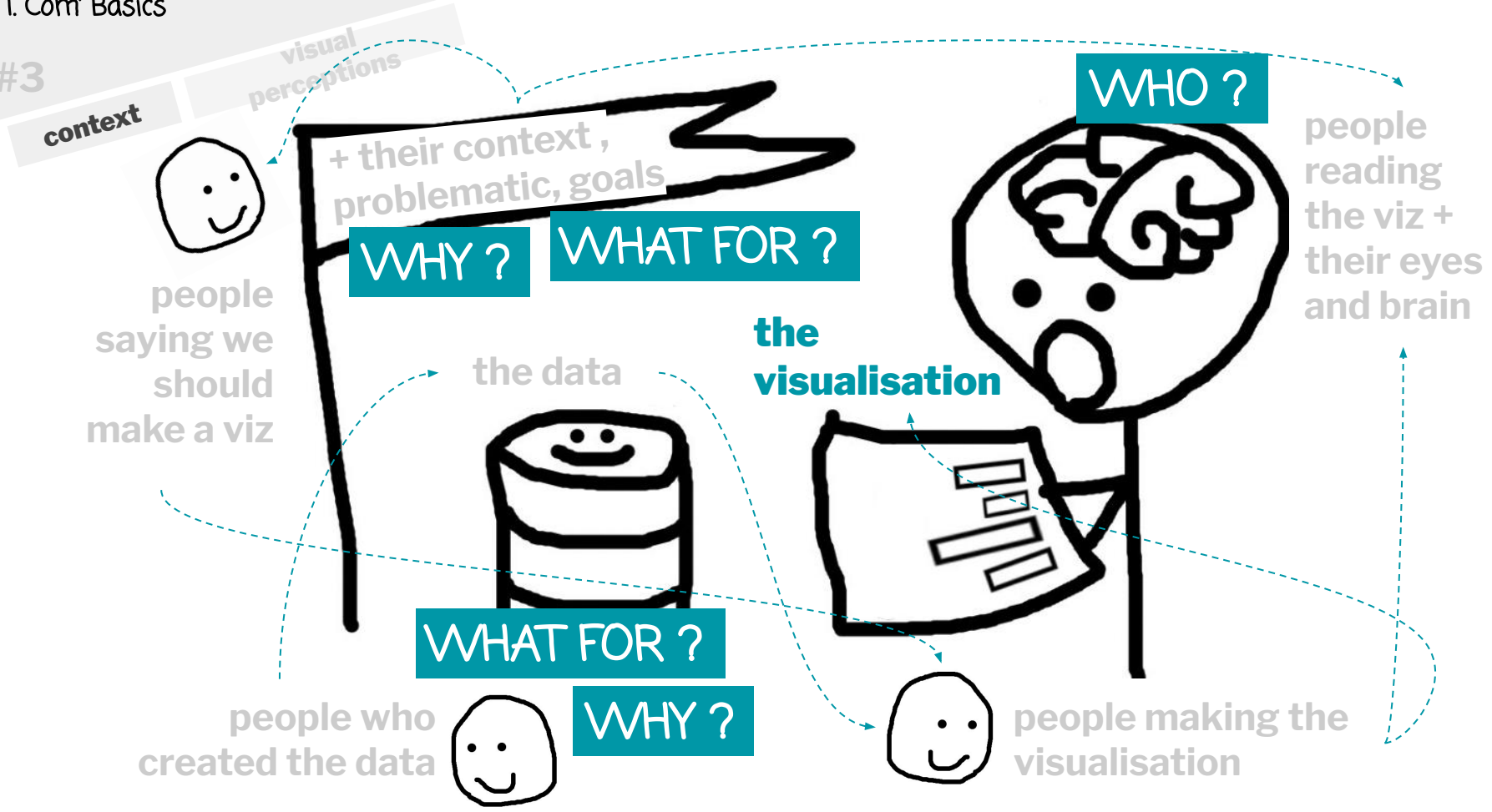
Three questions for creating a chart

1. What's your point? Headline	HYPOTHESIS
2. How can you emphasize your point in your chart? Chart type, data, color, highlights, annotations	PROOF
3. What does the final chart show exactly? Description, legend/keys, source	EXPLAINING



- GOOD TITLE
- ANNOTATE
- EXPLAIN







What **context** are you working in ? Any interests at stakes ?

What are your **goals** :
Attention / Understanding /
Implication

Who is your **audience**
their literacy and learning
types, their needs

What's the main
message of your chart
? Can you see it in the title ?
Do you provide a bit of
context ?

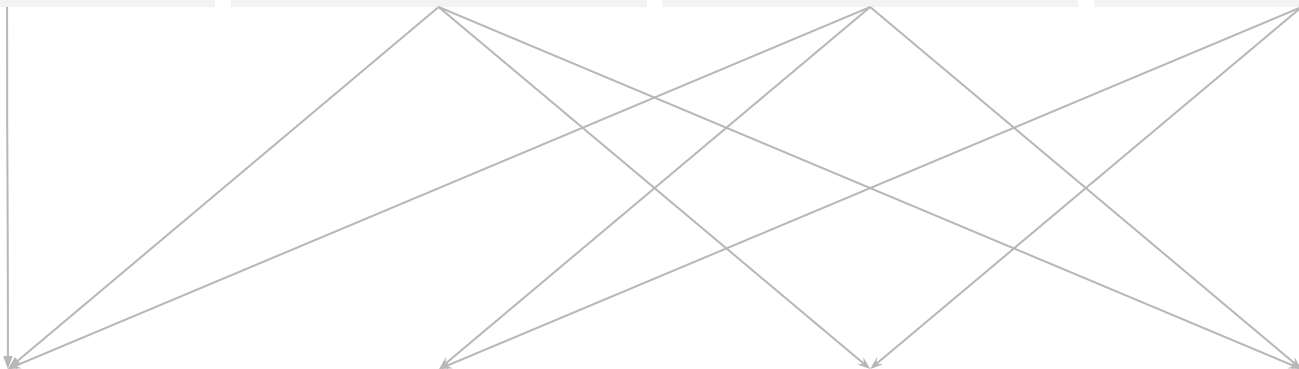


What is the **type** of your
visual : Declarative <=>
Exploratory

What are the
function(s) of your
visualisation (the chosen
chart type)

What are the **visual
variable** you use ?

How can you **explain,
annotate,
highlight** your chart ?



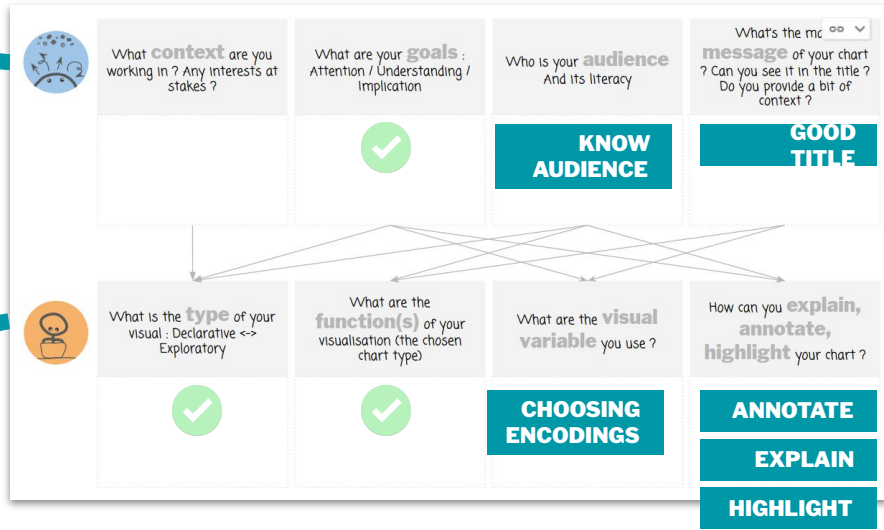
best
practices

1

2

3

visual
principles



LEGEND
CLOSE TO DATA

STRUCTURE
INFORMATION

USE
PADDING

CONSISTENT
COLOR CODING

LESS
IS MORE

more
best
practices

Thoughts & How To's / May 29, 2018

What to consider when choosing colors for data visualization

Thoughts & How To's / Sep 4, 2020

How to pick more beautiful colors for your data visualizations

Thoughts & How To's / Jun 23, 2020

What to consider when visualizing data for colorblind readers

Thoughts & How To's / Mar 21, 2018

What to consider when creating choropleth maps

Thoughts & How To's / Dec 13, 2019

The case for choosing clarity over complexity in data visualizations

Thoughts & How To's / May 21, 2019

What to consider when creating tables

Thoughts & How To's / Jan 3, 2018

What to consider when creating pie charts

Thoughts & How To's / Jan 9, 2018

What to consider when creating area charts

Core Principles of Data Visualization

PolicyViz

Data Visualization Checklist

by Stephanie Evergreen & Ann K. Emery
February 2019

This checklist is meant to be used as a guide for the development of high impact data visualizations. Rate each aspect of the data visualization by circling the most appropriate number, where 2 points means the guideline was fully met, 1 means it was partially met, and 0 means it was not met at all. Not all guidelines should be used frequently, but reserved for when the guideline truly does not apply. For example, a pie chart has no axis lines or tick marks to rate. If the guidelines have been broken intentionally to make a point, rate a 0 and deduct those points from the total possible. Refer to the Data Visualization Guidelines Chart on the last page for guidance on vocabulary and the Resources at the end for more details.

Text	Guideline	Rating
Graphs don't contain much text, so existing text must encapsulate your message and guide a viewer.	6-12 word descriptive title is left justified in upper left corner. Short titles enable readers to comprehend takeaway messages even while quickly scanning the graph. Rather than a generic phrase, use a descriptive sentence that encapsulates the graph's finding (e.g. "Why? Western cultures start reading in the upper left, so locate the title there.")	2 1 0 NA
Subtitles and/or annotations provide additional information.	Subtitles and/or annotations (call out text within the graph) can add explanatory and interpretive power to a graph. Use them to answer questions a viewer might have or to highlight specific data points.	2 1 0 NA
Text size is hierarchical and readable.	Titles are a larger size than subtitles or annotations, which are larger than axis labels, which are larger than source information. The smallest text - axis labels - are at least 9 point font size on paper, at least 20 on screen.	2 1 0 NA
Text is horizontal.	Titles, subtitles, annotations, and data labels are horizontal (not vertical or diagonal). Line labels and axis labels can deviate from this rule and still remain full panels. Consider anchoring graphs to orientation (e.g. from columns to bar charts) to make text horizontal.	2 1 0 NA
Data are labeled directly.	Position data labels near the data rather than in a separate legend (e.g. on top of or next to bars and next to lines). Extra horizontal legends are possible because our eyes move back and forth between the legend and the data can interrupt the brain's attempts to interpret the graph.	2 1 0 NA
Labels are used sparingly.	Focus attention by removing the redundancy. For example, in line charts, label every other year on an axis. Do not add numeric labels "and" use a y-axis scale, since this is redundant.	2 1 0 NA